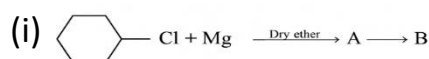


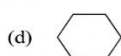
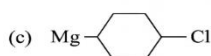
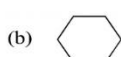
CHEMISTRY BOARD QUESTION PAPER: JULY 2022**Time: 3 Hrs. Max. Marks: 70****General Instructions:**

The question paper is divided into four sections.

- (1) Section A: Q. No. 1 contains Ten multiple choice type of questions carrying One mark each. Q. No. 2 contains Eight very short answer type of questions carrying One mark each.
- (2) Section B: Q. No. 3 to Q. No. 14 are Twelve short answer type of questions carrying Two marks each. (Attempt any Eight)
- (3) Section C: Q. No. 15 to Q. No. 26 are Twelve short answer type of questions carrying Three marks each. (Attempt any Eight)
- (4) Section D: Q. No. 27 to Q. No. 31 are Five long answer type of questions carrying Four marks each. (Attempt any Three)
- (5) Use of log table is allowed. Use of calculator is not allowed.
- (6) Figures to the right indicate full marks.
- (7) For each multiple choice type of question, it is mandatory to write the correct answer along with its alphabet. Only first attempt will be considered.

SECTION – A**Q.1. Select and write the correct answer: [10]**

the product 'B' is _____.



(ii) General electronic configuration of 3d series of d-block elements is _____.

- (a) $[\text{Ar}]3d^{1-10} 4s^2$ (b) $[\text{Xe}]3d^{1-10} 6s^2$ (c) $[\text{Kr}]3d^{1-10} 5s^2$ (d) $[\text{Rn}]3d^{1-10} 7s^2$

(iii) Correct IUPAC name of tert-butyl alcohol is _____.

- (a) 2-Methylbutan-1-ol (b) 2-Methylbutan-2-ol (c) 2-Methylpropan-2-ol
(d) 2-Methylpropan-1-ol

(iv) Standard emf at 298K: $\text{Zn(s)} | \text{Zn}^{2+}(1\text{M}) || \text{Cr}^{3+}(0.1\text{M}) | \text{Cr(s)}$. $E^\circ_{\text{Zn}} = -0.76\text{V}$, $E^\circ_{\text{Cr}} = -0.74\text{V}$.

- (a) -0.02V (b) $+0.02\text{V}$ (c) -0.2V (d) $+0.2\text{V}$



(v) Chlorine has +7 oxidation state in:

(a) HOCl (b) HClO₂ (c) HClO₃ (d) HClO₄

(vi) Work done during irreversible expansion (2 mol He, 2 dm³ → 4 dm³, 1 bar, 298K) is _____.

(a) 2.0 kJ (b) -2.0 kJ (c) 0.2 kJ (d) -0.2 kJ

(vii) Relation between edge length & radius in simple cubic lattice:

(a) 2a=r (b) 3a=4r (c) a=2r (d) 2a=4r

(viii) Lactose on hydrolysis gives _____.

(a) Galactose + Glucose (b) 2 Glucose (c) Fructose + Glucose (d) Fructose + Galactose

(ix) ZWT in green chemistry stands for _____.

(a) Zero waiting time (b) Zero waste technology (c) Zubl water technology

(d) Zhen wu tang

(x) Most basic amine:

(a) CH₃NH₂ (b) (CH₃)₂NH (c) (CH₃)₃N (d) C₂H₅NH₂

Q.2. Answer the following: [8]

(i) Relation between molar conductivity & conductivity.

(ii) Effective atomic number of Co³⁺ in [Co(NH₃)₆]³⁺.

(iii) Reaction converting phenol → salicylic acid.

(iv) IUPAC name of α-methylpropionic acid.

(v) Formula of Hinsberg's reagent.

(vi) Monomer of Nylon-6.

(vii) Cell representation of SHE.

(viii) Chemical composition of Ziegler-Natta catalyst.

SECTION – B

(Attempt any Eight) [16]

Q.3. Define: (i) Osmotic pressure (ii) Ebullioscopic constant

Q.4. pH = 3.12 → calculate [H₃O⁺].

Q.5. Kohlrausch Law & one application.

Q.6. Schottky defect vs Frenkel defect.



- Q.7. Derive $\Delta H - \Delta U$ relationship for gas phase reactions.
Q.8. Action of chlorine on: (i) NH_3 (excess) (ii) Phosphorus
Q.9. Molecular formula of: (i) Chalcopyrite (ii) Calamine
Q.10. Show that $t(99.9\%) = 3 \times t(90\%)$.

Q.11. Convert ethyl bromide $\rightarrow$ (i) Ethyl iodide (ii) Ethyl fluoride
Q.12. Explain linkage isomerism with example.
Q.13. Action of: (i) SOCl_2 (ii) P_2O_5 on carboxylic acids
Q.14. Reactions of carbolic acid with: (i) Br_2 water (ii) Conc. HNO_3

SECTION – C

(Attempt any Eight) [24]

- Q.15. Note on aldol condensation.
Q.16. Lanthanoid contraction; similarities between lanthanoids & actinoids.
Q.17. Calculate ΔH_f of CH_3OH using given data.
Q.18. What happens when: (i) Ethene + ICl (ii) $\text{SO}_2 + \text{V}_2\text{O}_5$ (iii) $\text{Cu} + \text{conc. H}_2\text{SO}_4$
Q.19. Number of atoms & unit cells in 0.5 g Nb (BCC); preparation of glucose from sucrose.
Q.20. Define nanochemistry; reaction of 1° & 2° alcohol vapours over hot Cu.
Q.21. FP depression problem; reduction of R-NO_2 (Sn/HCl).
Q.22. Denaturation of protein; derive Ostwald's dilution law for weak acid.
Q.23. Define nanotechnology; two applications of electrochemical series.
Q.24. Mechanism steps $\text{NO}_2 + \text{F}_2 \rightarrow \dots$; overall reaction, rate law, intermediate; preparation of Teflon.
Q.25. Define elastomer; two postulates of Werner's theory.
Q.26. Four salient features of SN_1 mechanism; carbylamine test.

SECTION – D

(Attempt any Three) [12]

- Q.27. Boiling point elevation problem (ethyl acetate); pseudo first-order reaction.
Q.28. Why aq. CuSO_4 turns blue litmus red? Why transition metal compounds are coloured?
Q.29. Hess's Law; interhalogen compounds; two uses of neon.
Q.30. Homoleptic vs heteroleptic complexes; convert carboxylic acid $\rightarrow$ (i) Ester (ii) Amide
Q.31. Define green chemistry; complete reaction ($\text{Isopropyl alcohol} \rightarrow \text{A} \rightarrow \text{B}$); action of hot HI on glucose.

